

Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Some things are discovered by accident!



Learning Outcome

Exchange information about unlikely events or fairly certain conditions that will be fulfilled in present, past or future in spoken or written texts and in work and social contexts.

Can Do Statements

- I can talk about science, technology and gadgets.
- I can express that an action is dependent on another.
- I can talk about two actions together.

Ask your partner:

- Have you ever found out something by accident? What? When?
- In what types of professions are people most likely to do this?

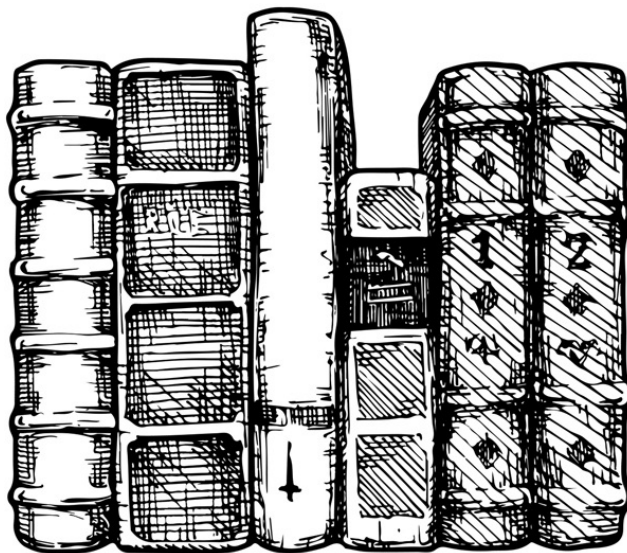




Let's Explore!

You see **dancing lights*** on the table!

* When "*lights dance*", they move around quickly on an object, not following a pattern.

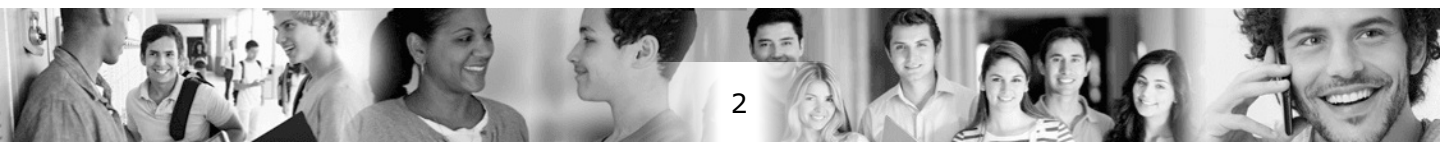


Instructions: Read the text aloud.

If you haven't heard of Wilhelm Conrad Röntgen, I'm telling you about him now. He was working in his laboratory expecting to find one thing when he discovered another. Let's go back to that day ... He's going to cover an electric bulb with black paper for an experiment if he hasn't already. There. Now if he's switched it on, you can see dancing lights on the table. If you've looked closely, you're seeing the light reflecting from a piece of paper covered with metal salts!

If you've seen the professor hold the paper up to the lamp and placed a book, a pack of cards, a piece of wood, and a door key between it and the lamp, you'll be aware that the rays from the lamp went through everything but the key. Now he has asked his wife to place her hand on a photographic plate. She agrees standing still is best, and he shines the black lamp on her hand for several minutes. Imagine watching the photograph develop, showing only the bones of her hand and a ring on her finger! If you've guessed he's discovering "the Unknown" or X-ray, you'd be right!

I admit liking to read about such historic things. I do it all the time. I enjoy finding out how things come to be. When a question comes up I am forced to look it up online. Sometimes it's challenging to find an answer. When this happens I stop in at the library and haul out the old tomes that keep such things for us. I'll search for anything if I've got time. If I've failed to find the answer in one publication, I'm soon moving to the next. I hope to keep unearthing new and intriguing facts. I'll invite you along next time if you've found this as stimulating as I do.





Let's Practice!

Have you heard of Wilhelm Conrad Röntgen?

Instructions: Read the sentences. Circle the correct answer to correctly complete the sentences.

1. He (working / was working / was work) expecting to find one thing when he discovered another.
2. I admit liking to (read / like to read / likes to read) about historical things.
3. I enjoy (find out / find out / finding out) how things came to be.
4. Sometimes it's (challenging to find / challenge to find / challenging to finding) an answer.
5. When a question comes up I am (forcing to look it up / forced to look it up / forced look it up) online.



Let's Practice!

Let's learn some useful ways to describe science, technology and gadgets.

Instructions: Make a line to match the word and its definition.

- | | |
|-------------------|--|
| 1. to haul | a. not moving |
| 2. to unearth | b. to pull something heavy |
| 3. tome | c. to go in one side and out the other |
| 4. to stand still | d. a large, heavy book |
| 5. to go through | e. to discover information |





Extension Activity

Read up on accidental discoveries.

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Go online or to the library and find out about 3 accidental discoveries that interest you.
- Find out the who, what, where, why, when and how of the discoveries.
- Write a brief summary for each and practice talking about them.
- Present your accidental discoveries to a partner or group.
- Listen to other discoveries.
- Discuss which you think are the most surprising and why.

Self-Assessment

Yes No

I find out about 3 accidental discoveries.

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I summarize the who, what, where, why, when and how of the discoveries.

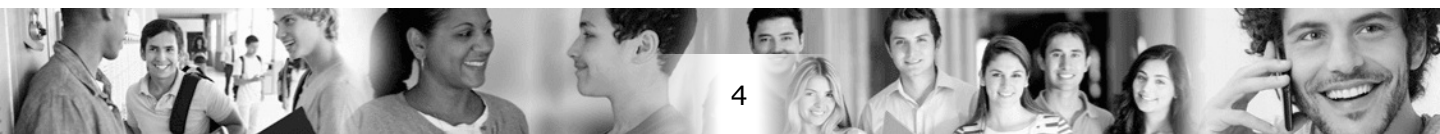
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I present my discoveries and discuss them.

☐ ☐


Target Words

- laboratory** (noun): a room for scientific research
e.g. He was working in his laboratory expecting to find one thing when he discovered another.
- metal salts** (noun): salts produced by immersing metal in acid
e.g. If you've looked closely, you're seeing the lights reflecting from a piece of paper covered with metal salts!
- to go through** (verb): to go in one side and out the other
e.g. The ray went through everything but the key.
- to stand still** (verb): to not move
e.g. She agrees to stand still, and he shines the black lamp on her hand for several minutes.
- to develop** (verb): to treat with chemicals to make a photograph
e.g. Imagine watching the photograph develop, showing only the bones of her hand and a ring on her finger!
- x-ray** (noun): radiation that can go through solid substances
e.g. If you've guessed he's discovering "the Unknown" or X-ray, you'd be right!
- to haul** (verb): to pull something heavy
e.g. When this happens I stop in at the library and haul out the old tomes that keep such things for us.
- tome** (noun): a large, heavy book
e.g. When this happens I stop in at the library and haul out the old tomes that keep such things for us.
- to unearth** (verb): to discover information after careful searching
e.g. I hope to keep unearthing new and intriguing facts.





Learning Points

The Real Conditional: Conditional Connections

The Real Conditional can be used to connect actions and consequences together. You can talk about possible future event with a Present Perfect verb in the if-clause and a future form in the main clause.

You use the Real Conditional to talk about events that are likely to happen if a condition is met, or situations that are always true if an action is or isn't completed.

You can use a verb in the Present Perfect with the if-clause to suggest that the action needs to be completed.

Structure:

If + Present/Present Perfect ... Future form (will, present continuous, be going to)

Example:

If I've failed to find the answer in one publication, I'm soon moving to the next.

Examples:

- *If you haven't heard of Wilhelm Conrad Röntgen, I'm telling you about him now.*
- *He's going to cover an electric bulb with black paper for an experiment if he hasn't already.*
- *Now if he's switched it on, you can see dancing lights on the table.*
- *If you've looked closely, you're seeing the light reflecting from a piece of paper covered with metal salts!*





Learning Points

Verbs Followed by -ing and To-Infinitives

When you talk about two actions linked together, you normally use two verbs. The first verb follows the rules of conjugation for the sentence, while the second verb can be a verb in the -ing form or to + Infinitive.

You can use two verbs to talk about two connected actions. The main verb introduces the main action, and the second verb in the -ing form gives more information about that action.

Verbs followed by -ing are *imagine, enjoy, consider, mind, keep, love/hate*.

Verbs followed by to + Infinitive are *ask, promise, agree, expect*.

Some verbs can be followed by either an -ing verb or to + Infinitive like *like, love, hate, dislike, intend*.

Structure:

Subject + first verb + -ing form of the second verb/to + Infinitive

Look at some examples from the text:

Examples:

- He was working in his laboratory **expecting to find** one thing when when he discovered another.
- Now he has **asked his wife to place** her hand on a photographic plate.
- She **agrees to stand still**, and he shines the black lamp on her hand for several minutes.
- **Imagine watching** the photograph develop, showing only the bones of her hand and a ring on her finger!
- I **admit liking to read** about such historic things.

